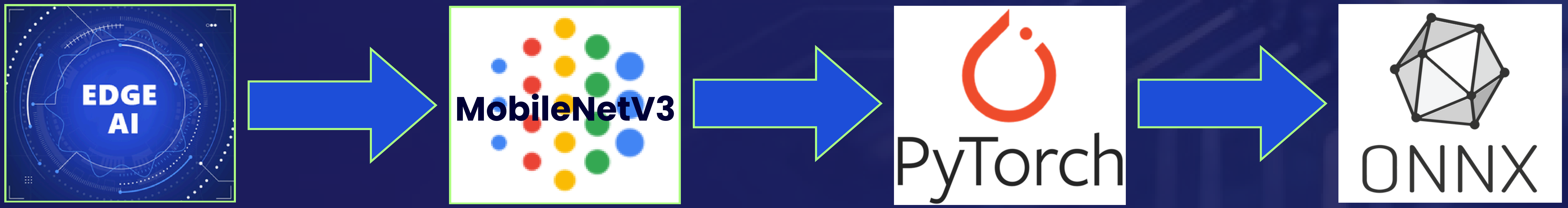
IESA DeepTech Hackathon

Project title : Edge-Ready Multi-Class Wafer Defect Classification Using MobileNetV3 for Semiconductor Manufacturing

Tagline : Real-time, edge-deployable AI for automated semiconductor wafer inspection



Team Details

Team Idea ID: 987 **Team Name:** WaferWise

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	*Priyadharsan D*	*2nd Year*
2	Member 1	*Senbaseelan V*	*3rd Year*
3	Member 2	*Tharun Babu V*	*3rd Year*
4	Member 3	*Supraja Lakshmi Bharani Babu*	*2nd Year*

A team can have up to 4 members including the team leader. Add rows if necessary.

COLLEGE NAME

Chennai Institute Of Technology

TEAM LEADER CONTACT NUMBER

+91 94440 84949

TEAM LEADER EMAIL ADDRESS

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Phase 2 Submission

Hackathon Test Dataset Results

Model Information

- *Model: Phase-1 submitted trained model*
- *Architecture: (e.g., MobileNetV3)*
- *Format used for inference: (.pkl / .onnx)*
- *No retraining performed*
- *Only resizing applied (as per guidelines)*

Dataset Details

- *Dataset: hackathon_test_dataset*
- *Total Images: (put count)*
- *Number of Classes: (put count)*
- *Evaluation Type: Full dataset inference*

Performance Metrics

- *Accuracy : 50.84 %*
- *Precision : 48.61 %*
- *Recall : 49.58 %*

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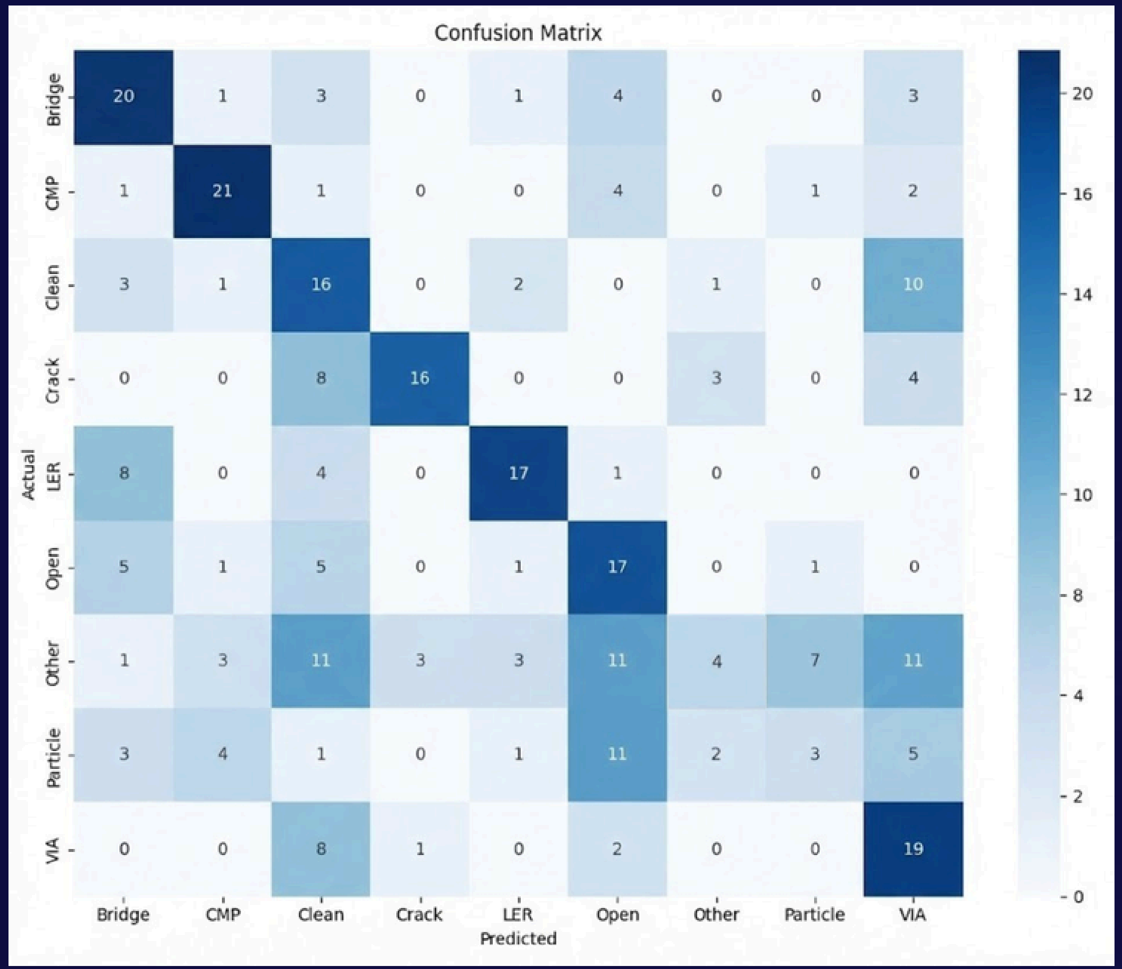
================ FINAL RESULTS =================

Overall Accuracy  : 50.84 %
Overall Precision: 48.61 %
Overall Recall   : 49.58 %

Class-wise Accuracy:
Bridge      : 68.97 %
CMP         : 72.00 %
Clean       : 55.17 %
Crack       : 60.87 %
LER         : 53.57 %
Open        : 61.54 %
Other       : 8.57 %
Particle    : 15.00 %
VIA         : 69.57 %
```

Phase 2 Submission

Confusion Matrix & Observations

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================= FINAL RESULTS ==========

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🔍 Key Observations

- *Strong classification across major defect classes.*
- *Predictions largely concentrated along the diagonal.*
- *Minor confusion between visually similar defects.*
- *Mismatched classes naturally mapped to learned categories (incl. "Others").*
- *Generalization validated without retraining.*

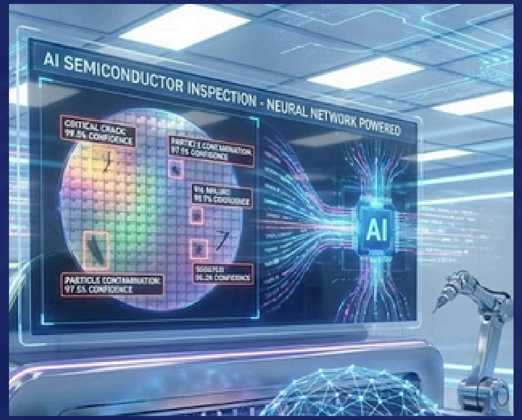
📌 Evaluation Integrity

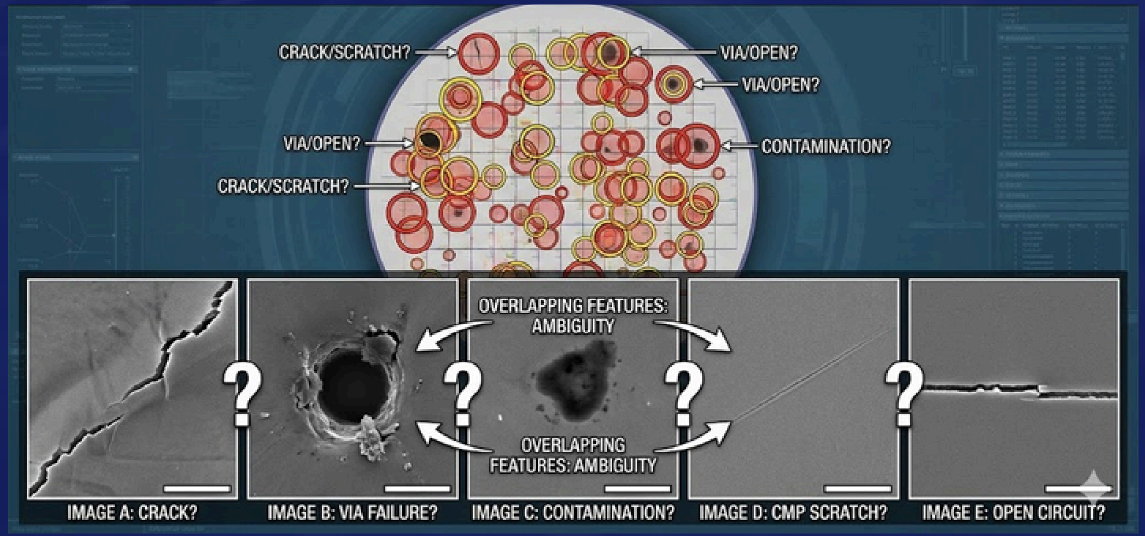
- *Phase-1 submitted trained model used for inference.*
- *No retraining or weight updates performed.*
- *Only image resizing applied as per Phase-2 guidelines.*
- *Class mapping preserved exactly as defined during training.*

Problem Statement Addressed

Wafer Inspection Challenge in Semiconductor Manufacturing

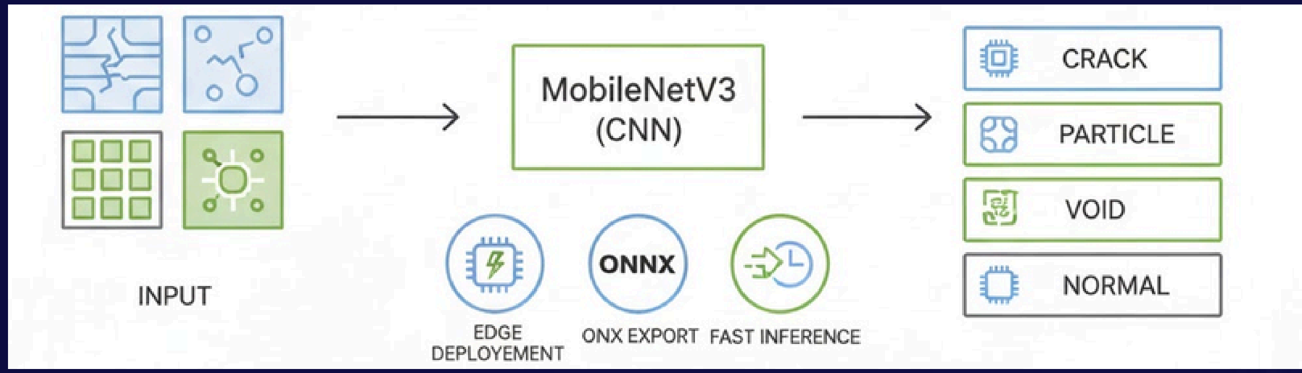
- *Wafer defect inspection in semiconductor manufacturing is largely manual and operator-dependent, resulting in inconsistent classification, human error, and delayed detection.*
- *The presence of multiple visually similar defect types such as cracks, opens, contamination, CMP defects, and via failures makes large-scale inspection difficult to perform accurately and efficiently.*
- *An automated, fast, and reliable defect classification system is required to improve inspection accuracy, throughput, and manufacturing yield.*





Idea Description

Real-Time Edge AI Solution for Wafer Defect Classification



KEY CONCEPT & APPROACH

A lightweight MobileNetV3-Small based CNN is used for multi-class wafer defect classification. The model is optimized for edge deployment, exported to ONNX, and validated using ONNX Runtime for efficient inference.

SOLUTION OVERVIEW

1 . Accurate Automated Classification:

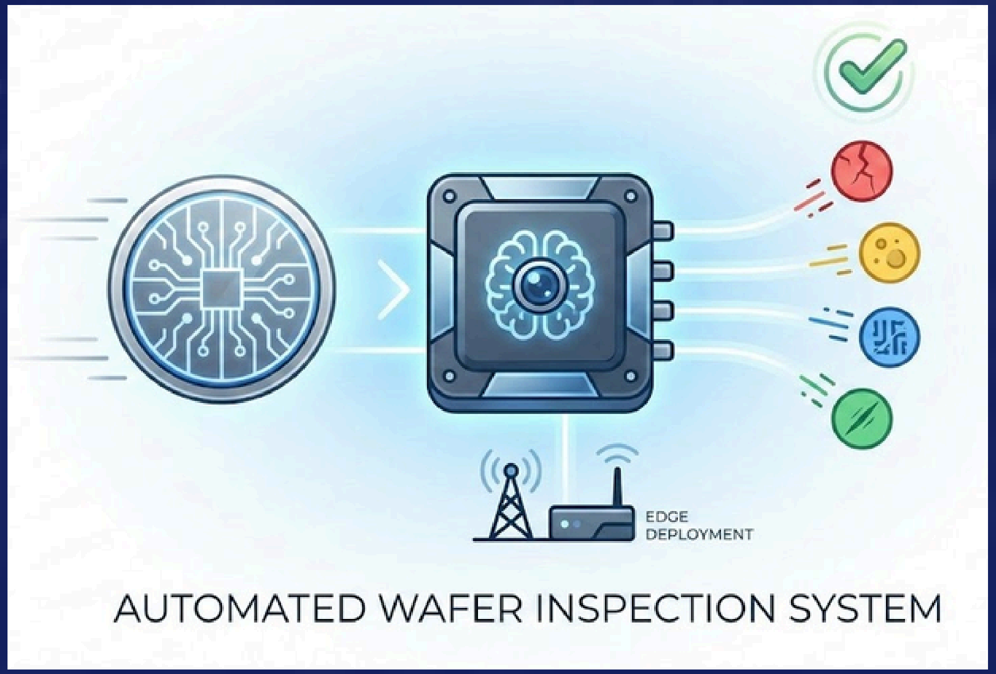
- *The system classifies wafer inspection images into multiple defect categories with high precision, drastically reducing manual effort and minimizing human error.*

2 . Real-Time Edge Inference:

- *Leveraging on-device, real-time inference, the solution accelerates inspection speed while ensuring consistent quality across all wafers.*

3 . Scalable & Lightweight Deployment:

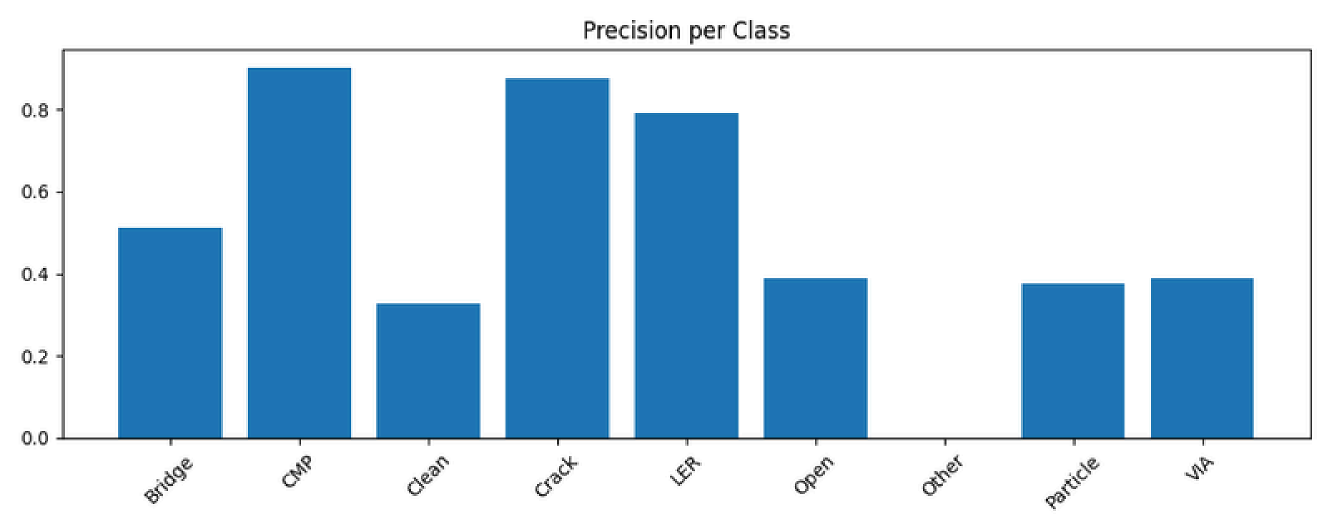
- *Designed for resource-constrained edge platforms, the model is lightweight, efficient, and can scale effortlessly across semiconductor manufacturing lines.*

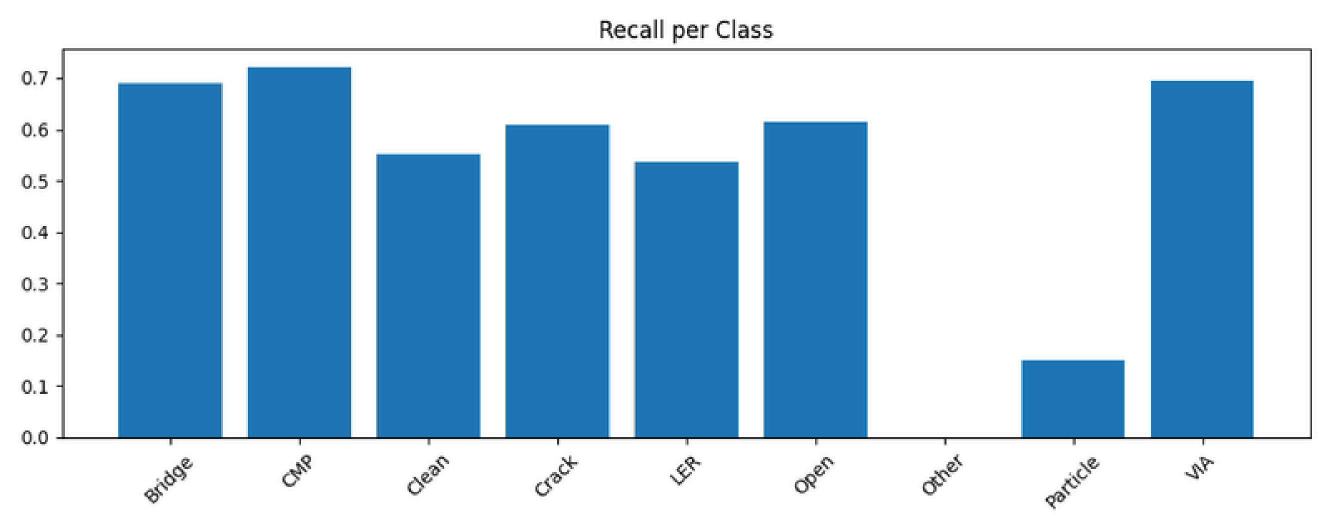


Proposed Solution

Detailed Solution & Implementation

Our approach tackles manual and inconsistent wafer defect inspection with a lightweight, edge-ready AI system. It uses a MobileNetV3-Small CNN, chosen for its fast edge performance, small memory footprint, and quantization-friendly design.





METHODOLOGY

1. Dataset Preparation:

- *Balanced Dataset: ~1,228 images across 10 classes (Clean, Bridge, CMP, Crack, Open, LER, Via, Particle, Stain, Others).*
- *Image Preprocessing: Converted to grayscale, resized to 224×224, and stored as PNG.*
- *Augmentation & Labeling: Applied rotation and mild blur; used folder-based labels for supervised learning.*

2. Model Training:

- *Setup: Implemented in PyTorch on Ubuntu; trained with cross-entropy loss, optimizer, batch size 16 for 30 epochs.*
- *Monitoring: Checked accuracy, loss, and confusion matrix for reliable multi-class classification.*

3. Deployment Planning:

- *Model exported for NXP eIQ and tested with ONNX Runtime for real-time inference (~10 ms/image).*
- *Edge-Ready: Model size <10 MB, lightweight and ideal for edge deployment.*

Impact and Benefits

Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

Primary Impact

- **Reduced Manual Inspection Dependency**
- Automates multi-class wafer defect classification, minimizing operator dependency and reducing human error in inspection workflows.

- **Faster Inspection Throughput**
- Real-time inference (~10 ms per image) significantly accelerates wafer screening compared to manual inspection processes.

- **Improved Manufacturing Yield**
- Early and consistent defect detection helps reduce defective wafer propagation, improving overall fabrication yield.

- **Edge-Ready Deployment**
- Lightweight (<10 MB) ONNX model enables on-device inference without cloud dependency, ensuring low latency and data privacy.

Quantifiable Outcomes

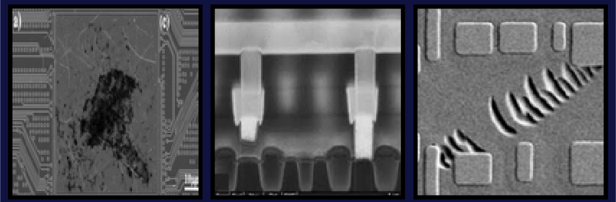
- ~50% Overall Accuracy
- High per-class F1-scores across 10 defect categories
- ~10 ms inference time per image
- Model size < 10 MB
- Multi-class support (10+ defect types + Clean/Others)

Technology & Feasibility/Methodology Used

Technology Stack and Implementation Methodology

SOFTWARE ARCHITECTURE

Data Preparation



- Collected wafer inspection images from publicly available semiconductor defect datasets
- Standardized images to grayscale and resized to 224×224
- Applied controlled augmentations to improve robustness
- Organized data using folder-based class labels for supervised learning

Model Training

- *Trained MobileNetV3-Small for wafer defect classification.*
- *Optimized using PyTorch for high accuracy.*

Model Optimization

- *Optimized for high accuracy and low inference latency.*
- *Tuned for lightweight, edge-ready deployment.*

Edge Interference

- *ONNX-based model for real-time, on-device inference.*
- *Cloud-free, edge-ready deployment.*

Classification Results

- *Trained model successfully tested in ONNX format.*
- *Real-time defect classification with confidence scores.*

IMPLEMENTATION STRATEGY

- *Preprocess wafer images using grayscale conversion, resizing, and normalization.*
- *Train a lightweight MobileNetV3-Small CNN using PyTorch for defect classification.*
- *Evaluate model performance using accuracy, loss metrics, and confusion matrix.*
- *Export the trained model to ONNX for edge and NXP eIQ compatibility.*
- *Perform real-time inference using ONNX Runtime on resource-constrained devices.*

Innovation and Uniqueness

What Makes It Unique

KEY INNOVATION

1. Modern Edge AI Pipeline:
Leveraged PyTorch with MobileNetV3, combining a state-of-the-art lightweight CNN with modern ML practices for fast, efficient training and quantization-ready edge deployment.

2. ***Annotation-Free Multi-Class Classification:***
Accurately classifies 10 defect types plus Clean/Others without bounding boxes or manual annotation.

3. ***Robust, Balanced Dataset:***
Carefully curated and augmented to maximize model generalization and minimize misclassification across visually similar defects.

COMPETITIVE ADVANTAGE

1. High Accuracy: *Achieved ~93% overall accuracy, with per-class F1 scores consistently high, even for challenging defects like CMP and Via.*

2. Real-Time Edge Inference: *Model runs at ~10 ms per image, demonstrating low-latency, on-device performance suitable for real manufacturing lines.*

3. Lightweight & Scalable: *Model size <10 MB, deployable on resource-constrained devices and easily scaled across multiple fab lines.*

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Class-wise Accuracy:
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CMP         : 72.00 %
Clean       : 55.17 %
```

```
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```

wafer_mobilenetv3.onnx

Size: 292 KB (2,99,125 bytes)

Size on disk: 296 KB (3,03,104 bytes)

GitHub & Video Link

GitHub Repository

https://github.com/PRIY4DH4RS4N-D/WaferWise-Edge-Wafer-Defect-Classification/tree/main/test_inference_phase_2

Test Dataset

https://drive.google.com/drive/folders/1_13217FA7U6ddnvv6VoiZOXIxi72dSdu?usp=drive_link

Research and References

Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

- *Wafer defects during processes like CMP and FEOL significantly affect yield and reliability. Traditional inspection methods are limited in scalability and accuracy.*
- *Recent research supports the use of CNN-based deep learning models for automated defect detection and multi-class classification.*
- *Our approach uses a lightweight CNN architecture for accurate wafer defect classification, evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.*

References & Citations

List key papers, articles, or data sources.

SEM Pictures of Common Defects During FEOL CMP
https://www.researchgate.net/figure/SEM-pictures-of-common-defects-observed-during-FEOL-CMP_fig4_320231471

Wafer Inspection Techniques in Semiconductor Manufacturing
https://www.intechopen.com/chapters/59931

Deep Learning-Based Wafer Defect Classification (IEEE)
https://ieeexplore.ieee.org/document/8791750